

Online Appendix

Dry Firms, Deep Recessions: Corporate Payouts and Aggregate Dynamics

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A Data Construction and Cleaning

A.1 Definition of liquid assets from the Financial Accounts

Following the definitions used in the Federal Reserve Board’s Financial Accounts, we measure liquid assets as the sum of private foreign deposits (FL103091003.Q), checkable deposits and currency (FL103020005.Q), total time and savings deposits (FL103030003.Q), and money market fund shares (FL103034000.Q). This is the Financial Accounts’ core measure of liquid assets and is the measure used throughout the paper.

For comparison and robustness, we also report a broader measure that additionally includes municipal securities (FL103062003.Q), commercial paper (FL103069100.Q), Treasury securities (FL103061103.Q), agency- and GSE-backed securities (FL103061703.Q), security repurchase agreements (FL102051003.Q), mutual fund shares (FL103064203.Q), corporate equities (FL103064103.Q), and corporate and foreign bonds (FL123063003.Q). Table A1 reports both measures relative to nominal GDP by decade.

Table A1: Corporate liquid assets relative to nominal GDP

Decade	<i>Decade averages</i>		Period	<i>Aggregated periods</i>	
	Core	Broad		Core	Broad
1970s	4.5%	9.6%	1970–1989	4.6%	10.2%
1980s	4.7%	10.9%			
1990s	4.9%	14.9%	1990–2009	6.3%	17.2%
2000s	7.6%	19.4%			
2010s	8.8%	19.4%	2010–present	10.3%	22.1%
2020s	12.8%	26.4%			

Notes: Values are quarterly averages, in percent. The current decade and the 2010–present period run through 2026:Q1.

A.2 Compustat Data Construction and Cleaning

We construct a panel of U.S. non-financial, non-utility firms from the Compustat North America annual fundamentals file. The main sample restrictions are as follows:

1. **Sample restrictions:** We retain only firm-year observations that satisfy the following:
 - Report in U.S. dollars (curcd = USD), use standard data format (datafmt = STD), industrial format (indfmt = INDL), consolidated accounting (consol = C), and the domestic population source (popsrc = D).

- Incorporated in the United States ($fic = USA$).
 - Have non-missing accounting dates ($datadate$).
2. **Firm-year identification:** Fiscal year ($fyear$) is the time identifier. We drop observations with missing fiscal year and observations flagged as duplicate $gvkey$ – $fyear$ records, leaving a panel that is unique at the firm-year level.
 3. **Industry exclusions:** We exclude firms in the financial and utility sectors, defined as those with NAICS codes beginning in 22 (utilities), 52 (finance), or 53 (real estate). Unclassifiable firms ($NAICS = 999$) are also excluded. Observations with missing or overly coarse (e.g., 2-digit) NAICS codes are dropped.
 4. **Data validity and trimming:** We apply the following filters to address data quality and outliers:
 - Sales must be strictly positive and total assets must exceed 1 in Compustat’s reported units.
 - The cash-to-assets ratio must lie within $[0, 1]$.
 - The cash-to-sales ratio must lie within $[0, 5]$.
 - Observations with negative debt (short or long term) are excluded.
 - Leverage (defined as total debt over assets) is trimmed at the 99th percentile.
 - Dividends-to-assets ratios exceeding 10 are excluded.
 - We trim sales observations below the 1st percentile or above the 99th percentile.

A.3 PSID Data Construction and Cleaning

Our empirical analysis draws on the biennial Panel Study of Income Dynamics (PSID) Family and Individual Files for survey years 2005 through 2021. We construct a household panel containing income, wealth, expenditure, and demographic variables. This appendix describes the variables and sample restrictions used in the regression analysis.

Data Sources and Structure

We extract a common set of variables from each Family File and append the survey-year files. We then merge them with the Individual File by interview identifier and year. The individual-file processing identifies the household head, head and spouse person numbers, and age-based dependent counts; only matched family–head records are retained.

Cleaning and Harmonization Procedures

- **Expenditure Variables:** The baseline regression outcome is the PSID Family File's reported total-expenditure measure. We also construct a component-sum measure from food, rent, utilities, health care, transportation, education, childcare, household durables, clothing, recreation, and related categories. Component amounts reported weekly, monthly, or at other frequencies are annualized.
- **Income and Dividend Variables:** We compute household-level dividend income as the sum of head and spouse self-reported income from dividends. Careful attention is given to avoid double-counting in cases where joint asset ownership is reported. We similarly construct interest, rent, and trust income, along with earned labor income and transfer payments.
- **Wealth Variables:** We define wealth as cash and savings plus stocks, bonds, net non-primary-residence real estate, net vehicle value, and the household's net share of business or farm assets. This measure excludes the primary residence and does not separately subtract mortgages or other debts.
- **Deflation:** We merge December observations of the seasonally adjusted CPI-U series CPIAUCSL. Dividend income, the constructed component-sum consumption measure, wealth, and labor-income controls are divided by the contemporaneous December CPI and multiplied by 100. The reported total-expenditure outcome is divided by the following December's CPI because of the expenditure variable's survey timing. These transformations express values in CPI-index-base dollars.
- **Sample Restrictions:** The merge retains records with an identified household head that match a Family File observation. The regressions use complete cases for the dependent variable, dividend income, and the controls included in each specification.

Key Variables for Analysis

The core regression examines the relationship between *real dividend income* and *real household expenditure*. The key variables used include:

- `divtotal_real`: Sum of head and spouse dividend income, deflated.
- `totexpenditure_real`: PSID-reported total household expenditure, deflated using the following December's CPI.

- `wealth_real`, `hlabinc_real`, `wlabinc_real`: Real wealth and labor income of household members.
- `id`, `year`: Household and year identifiers used for fixed effects.

A.4 Details on micro-fit moments

This appendix evaluates the calibrated model against firm-level moments that are central to the cash-hoarding mechanism but are not directly targeted in the calibration. The exercises discipline three distinct objects: the cross-sectional distribution of cash holdings, the dividend-smoothing role of cash, and the marginal propensity to pay out (MPPO) after an exogenous liquidity shock.

Cash holdings across the firm distribution

The model predicts that the dispersion of cash holdings should decline with firm size or productivity. The reason is that low-productivity firms have high precautionary demand but limited internal resources, while high-productivity firms have enough internal funds to reach their target cash position. This non-monotone cash policy implies that cash-to-sales and cash-to-assets are more dispersed at the bottom of the firm distribution than at the top.

We test this prediction in Compustat by computing the standard deviation of cash ratios across percentiles of the sales distribution. Figure A1 implements this comparison directly. In the data, we residualize cash ratios using

$$n_{it} = \Gamma X_{it-1} + \alpha_i + \lambda_{st} + \varepsilon_{it},$$

where n_{it} is cash-to-sales, X_{it-1} includes lagged size, leverage, and sales growth, α_i are firm fixed effects, and λ_{st} are SIC-code-by-year fixed effects. We compute the standard deviation of the residual cash ratio within each percentile of the sales distribution. The model panel reports the corresponding model-implied cash-to-sales dispersion across productivity states. In both, dispersion is largest at low sales or productivity and declines along the firm distribution.

Figure A2 shows that the same pattern is visible in levels, not only in dispersion. Average cash-to-sales declines with sales in Compustat, and the model generates the analogous negative relationship between cash-to-output and output. The two figures therefore validate both parts of the distributional prediction: small firms have higher and more dispersed liquidity ratios, while larger firms are closer to satiation.

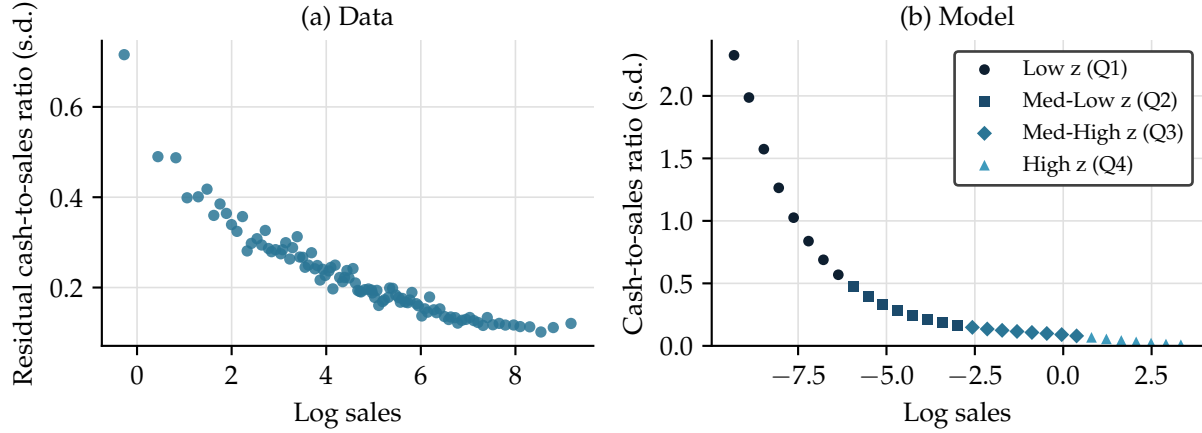


Figure A1: Dispersion of cash holdings across the firm distribution

Notes: Panel (a) plots the standard deviation of residual cash-to-sales ratios within percentiles of the sales distribution in Compustat. Panel (b) plots the corresponding model-implied cash-to-sales dispersion across productivity states.

Cash holdings and dividend smoothing

The second prediction is that cash buffers allow firms to smooth dividends in downturns. We test this mechanism with annual Compustat data from 1981 to 2019 using the local projection

$$d_{it+h} = \beta_h \text{Rec}_t + \delta_h n_{it-1} + \gamma_h \text{Rec}_t \times n_{it-1} + \Gamma_h X_{it-1} + \alpha_i + \varepsilon_{it+h},$$

where d_{it+h} is log real dividends, Rec_t indicates a year of negative real GDP growth, n_{it-1} is lagged cash-to-assets, and X_{it-1} includes lagged size, leverage, and sales growth.

Figure A3 shows that dividends fall in recessions, but firms with higher lagged cash reduce payouts by less. The left panel measures the estimated recession coefficient when lagged cash is zero, while the right panel measures how that coefficient changes with a one-percentage-point increase in lagged cash-to-assets. The interaction is positive on impact and one year ahead, although the impact estimate is small and imprecise; the one-year-ahead interaction is positive and statistically significant at the 10 percent level. Quantitatively, a one standard-deviation increase in cash holdings reduces the recession dividend drop by roughly 50 percent.

Figure A4 repeats the same regression in simulated model data. The model reproduces the two key qualitative patterns: dividends decline in bad aggregate states, and the decline is muted when firms enter the state with more cash. In particular, a one standard deviation increase in cash holdings also reduces the dividend drop by 50%, quantita-

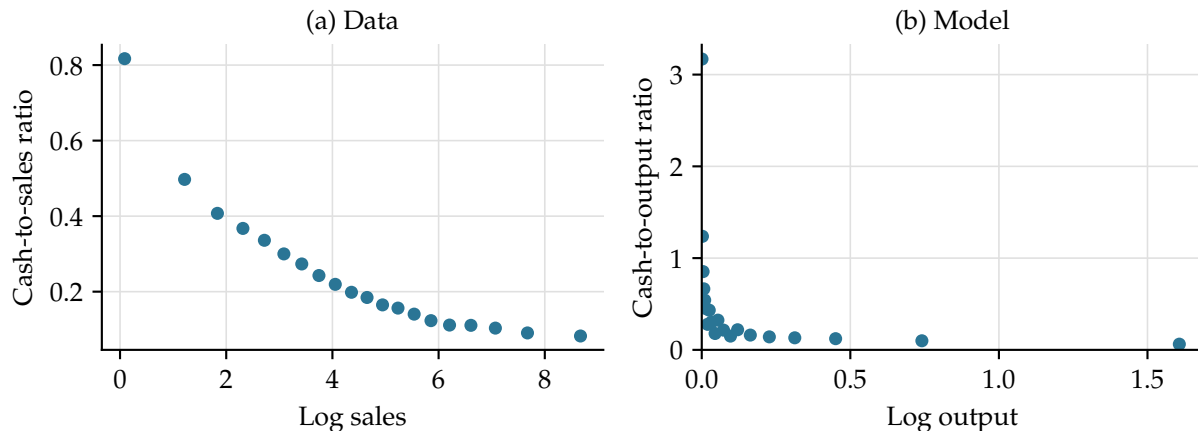


Figure A2: Average cash holdings across the firm distribution

Notes: The figure compares the average cash-to-output ratio over the output distribution in the model with the average cash-to-sales ratio over the sales distribution in Compustat.

tively similar to the empirical moment. This is important because the regression is not a calibration target; it is an untargeted validation that the dividend-smoothing mechanism survives aggregation and has a similar magnitude in the data and in the simulated economy.

We repeat the exercise using the log of one plus real dividends as the dependent variable, thereby retaining observations with zero dividends, and using total net payouts, defined as dividends plus buybacks minus equity issuance. Results are available upon request.

Quantitative MPPO validation using NOL carryback liquidity shocks

The third exercise confronts the model-implied marginal propensity to pay out (MPPO) with an empirical estimate of the same dollar-response object. In the model, the MPPO is the payout generated by an additional dollar of firm liquidity and can be computed directly across output states. In the data, we estimate it from plausibly exogenous liquidity shocks and sort firms by lagged operating profits, an observable proxy for their position along the output or productivity distribution. This produces a quantitative comparison between the model's MPPO profile and its empirical counterpart.

We use the 2002 and 2009 expansions of net operating loss (NOL) carrybacks as liquidity shocks. Under the old rule, firms could carry current losses back two years and recover taxes previously paid. The reforms expanded the carryback window to five years for losses in 2001–2002 and 2008–2009, allowing eligible firms to recover taxes paid in

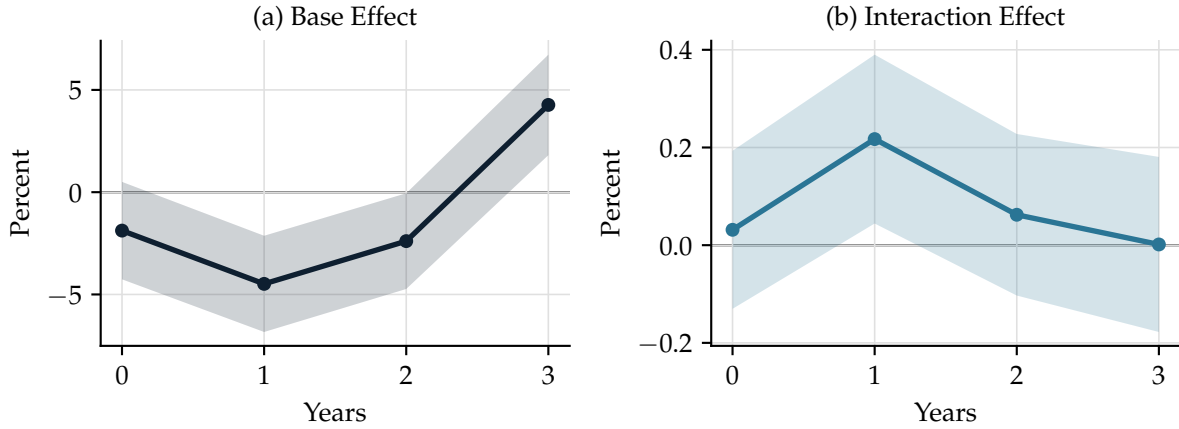


Figure A3: Dividend smoothing by cash holdings in Compustat

Notes: The figure reports the recession effect on dividends and the interaction between recessions and lagged cash holdings in Compustat. Shaded bands are 90 percent confidence intervals.

more distant profitable years.

The timing of the two reforms guides how we assign the predicted refund to payout decisions. The 2002 expansion was enacted early in 2002 and applied to NOLs from taxable years ending in 2001 and 2002. Firms could therefore receive, or at least anticipate receiving, refunds associated with both loss years when setting payout policy during 2002. We assign the sum of the incremental predicted refunds from the 2001 and 2002 loss years to the 2002 payout event. The 2009 expansion was enacted late in 2009 and allowed firms to elect the extended carryback for one eligible loss year, either 2008 or 2009. We therefore assign to the 2009 payout event the larger incremental predicted refund generated by the 2008 and 2009 loss-year calculations. In both cases, the impact specification uses payout in the event year, while the one-year-ahead specification allows for delayed payout adjustment.

For each eligible loss year ℓ , we define the reform-induced predicted refund as

$$\Delta Refund_{i\ell} = \min \{ \tau NOL_{i\ell}, Capacity_{i\ell}^{new} \} - \min \{ \tau NOL_{i\ell}, Capacity_{i\ell}^{old} \},$$

where we set $\tau = 0.35$, $Capacity_{i\ell}^{old}$ is positive federal income taxes reported over the previous two years, and $Capacity_{i\ell}^{new}$ is tax capacity under the expanded window. For the 2002 reform, we add the preceding five years of positive federal taxes at full weight. For the 2009 reform, the fifth preceding year enters at one-half weight, reflecting the WHBAA limitation. Missing or negative reported federal taxes are set to zero for this calculation. The minimum operator reflects that the refund cannot exceed either the tax value of the

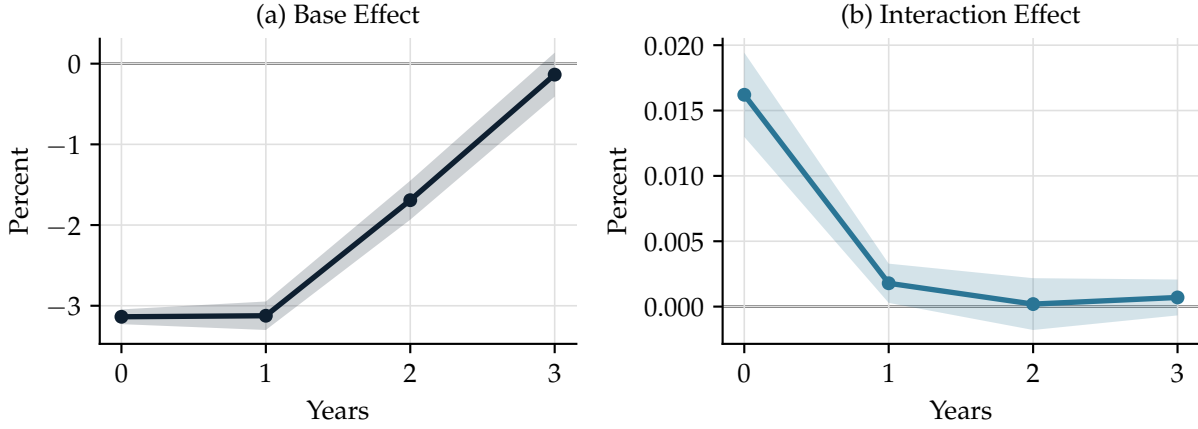


Figure A4: Dividend smoothing by cash holdings in the model

Notes: The figure repeats the local-projection exercise using model-simulated data. Shaded bands are 90 percent confidence intervals.

current loss or eligible prior tax payments. The event-level shock $\widehat{Refund}_{it}$ is the sum of $\Delta Refund_{i,2001}$ and $\Delta Refund_{i,2002}$ for the 2002 event, and the maximum of $\Delta Refund_{i,2008}$ and $\Delta Refund_{i,2009}$ for the 2009 event.

The identifying variation is the incremental liquidity created by the reform relative to the old two-year rule. The event sample contains firms with negative pretax income in the relevant event year; firms without positive incremental predicted refunds remain in the sample with a zero shock. Conditional on the current loss, contemporaneous firm conditions, and tax capacity under the old rule, variation in $\widehat{Refund}_{it}$ comes from taxes paid in years outside the old carryback window but inside the expanded window. This component is predetermined relative to current payout choices and became valuable only because of the policy expansion.

We estimate the MPPO by interacting the predicted refund with bins of lagged operating profits:

$$\frac{Payout_{i,t+h}}{A_{i,t-1}} = \sum_b \beta_{b,h} \left(\frac{\widehat{Refund}_{it}}{A_{i,t-1}} \times \mathbf{1}\{B_{i,t-1} = b\} \right) + \Gamma_h X_{i,t-1} + \Theta_h Z_{it} + \eta_b + \alpha_{s(i),t} + \varepsilon_{i,t+h}.$$

The outcome is total payout, defined as total dividends plus purchases of common and preferred stock minus issuance of common and preferred stock, scaled by lagged assets. $B_{i,t-1}$ denotes quartiles of lagged operating profits formed in the pooled two-event sample; these proxy for the firm's position in the output or productivity distribution. The vector $X_{i,t-1}$ includes lagged log assets, leverage, and sales growth. The vector Z_{it} in-

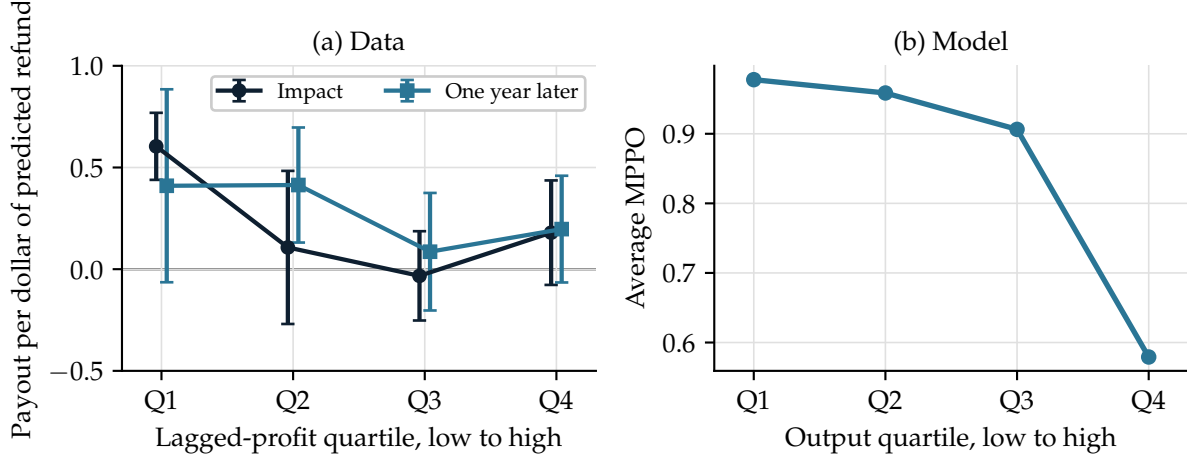


Figure A5: Marginal propensity to pay out across the firm distribution

Notes: Panel (a) plots the empirical payout response to one dollar of predicted NOL refund across quartiles of lagged operating profits. Coefficients are reported on impact and one year ahead; vertical bars are 90 percent confidence intervals. The empirical shock sums the 2001–2002 predicted refunds for the 2002 event and uses the larger of the 2008 and 2009 predicted refunds for the 2009 event. Panel (b) plots the model-implied average MPPO across output states. Bins move from lower to higher operating profits or output.

cludes the tax value of current losses and tax capacity under the old two-year carry-back rule. The fixed effects $\alpha_{s(i),t}$ are three-digit-NAICS-by-year fixed effects, and η_b are lagged-profit-bin fixed effects. Standard errors are clustered by firm. Before estimation, the predicted-refund-to-lagged-assets variable and the payout outcomes are winsorized at their 1st and 99th percentiles within the pooled event sample.

Because the refund and payout are scaled by the same predetermined variable, each coefficient $\beta_{b,h}$ is interpreted as dollars of payout per dollar of predicted refund for firms in bin b . We estimate the response on impact and one year ahead because the exact timing of refund receipt depends on filing and repayment lags.

Figure A5 provides the quantitative validation of the MPPO mechanism. In the model, average MPPO falls from nearly one at low output to about 0.6 at high output. In the data, the impact response is 0.60 dollars per dollar of predicted refund in the lowest profit quartile, compared with 0.11, -0.03 , and 0.18 in quartiles two through four. Thus the empirical profile features a sharp drop after the lowest-profit quartile, but is not strictly monotone across the remaining quartiles. We do not interpret the exercise as matching the level of MPPO in every bin. Instead, it disciplines the central shape restriction that pass-through is especially high among low-profit firms and substantially lower elsewhere in the operating-profit distribution.

The impact profile is statistically different from a flat MPPO schedule. In the pooled specification, equality between the lowest-profit quartile and each other quartile is re-

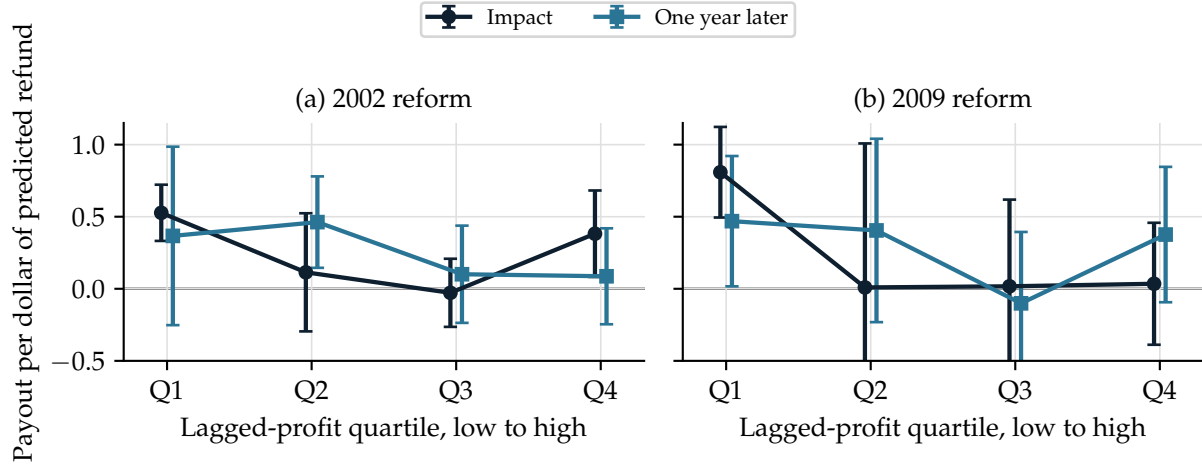


Figure A6: Empirical MPPO estimates separately by reform

Notes: The figure plots the payout response to one dollar of predicted refund across quartiles of lagged operating profits. The left panel uses the 2002 carryback expansion and the right panel uses the 2009 expansion. Coefficients are reported on impact and one year ahead; vertical bars report 90 percent confidence intervals.

jected, and joint equality of all four coefficients is rejected with $p < 0.001$, establishing substantial heterogeneity across quartiles. The pattern is absent in the year-minus-one check: the pairwise differences are small and joint equality is not rejected ($p = 0.471$). We report the reform-specific estimates, coefficient-difference tests, and pre-event check below as part of the same MPPO validation exercise.

Robustness and pre-event checks The first robustness check estimates the MPPO profile separately for the two policy episodes. This split is useful because the 2002 and 2009 reforms differ in timing and in the loss years that firms could use. In 2002, firms could benefit from losses around 2001–2002, while in 2009 firms could elect the extended carryback for one of the 2008 or 2009 loss years. Figure A6 shows that the declining profile is not driven mechanically by pooling the two reforms: in both episodes, the payout response is larger among firms in lower lagged-profit states than among firms in higher lagged-profit states.

Table A2 reports formal coefficient-difference tests. The pooled impact profile is statistically different across quartiles: the lowest-profit quartile differs from each other quartile individually, and equality of all four coefficients is rejected. The reform-specific tests point in the same direction, with significant differences between the lowest and third quartiles in 2002 and between the lowest and highest quartiles in 2009.

Figure A7 and Table A3 report the corresponding pre-event check. We repeat the same specification using payout in the year before the policy event as the dependent variable.

Table A2: Impact coefficient-difference tests

Sample	$p(Q_1 = Q_2)$	$p(Q_1 = Q_3)$	$p(Q_1 = Q_4)$	$p(Q_1 = Q_2 = Q_3 = Q_4)$
Pooled	0.045	< 0.001	0.021	< 0.001
2002 reform	0.135	0.003	0.495	0.024
2009 reform	0.209	0.054	0.018	0.058

Notes: Entries are p -values from tests of equality between quartile-specific MPPO coefficients. The pooled row uses both reforms. The 2002 and 2009 rows estimate the same specification separately within each reform episode.

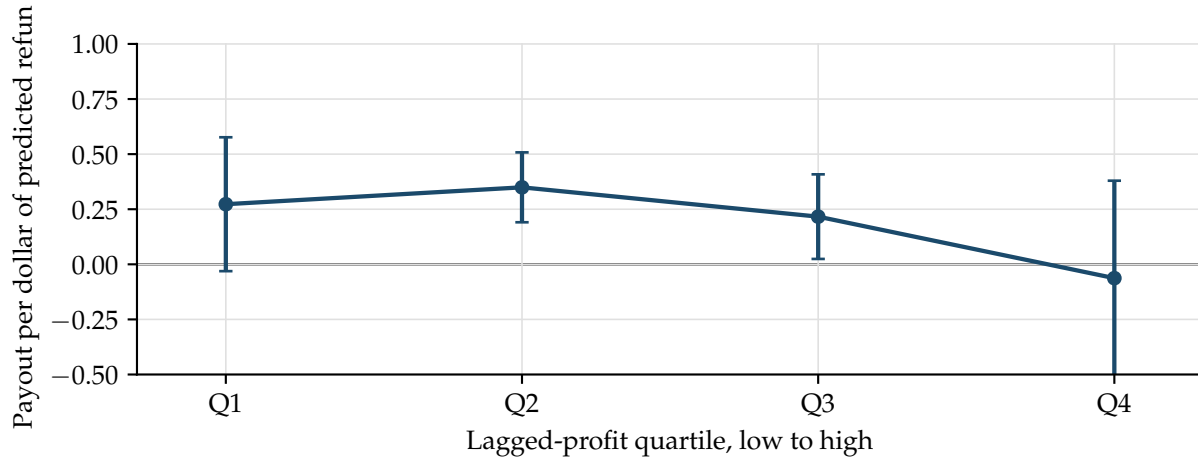


Figure A7: Year-minus-one payout response across lagged-profit quartiles

Notes: The figure plots the coefficient on the predicted refund interacted with lagged-profit quartiles when the dependent variable is payout in the year before the event. Vertical bars report 90 percent confidence intervals.

Table A3: Year-minus-one coefficient-difference tests

Sample	$p(Q_1 = Q_2)$	$p(Q_1 = Q_3)$	$p(Q_1 = Q_4)$	$p(Q_1 = Q_2 = Q_3 = Q_4)$
Pooled	0.712	0.795	0.298	0.471

Notes: Entries are p -values from the same coefficient-difference tests as in Table A2, but using payout in the year before the event as the dependent variable.

The shape of the impact MPPO profile is not present before the reform: the pooled year-minus-one coefficients are not significantly different from one another, either in pairwise comparisons with the lowest-profit quartile or in the joint equality test. This supports the interpretation that the post-reform differences capture differential payout responses to policy-induced liquidity rather than pre-existing payout trends across profit quartiles.

B Linear component of the equity issuance cost

This subsection adds a proportional component to the equity issuance cost in the two-period model. We use the notation of Section 2: period-2 variables are written as explicit functions of the productivity realization A_2 , rather than with tildes. The cost is

$$C(d) = \begin{cases} -\mu_1 d + \frac{\mu_2}{2} d^2, & d < 0, \\ 0, & d \geq 0, \end{cases} \quad \mu_1 \geq 0, \quad \mu_2 > 0.$$

Because equity issuance corresponds to $d < 0$, the term $-\mu_1 d$ is a positive proportional cost.

The firm enters period 1 with cash n_1 , observes A_1 , and chooses next-period cash $n_2 \geq 0$. Net payouts are residual:

$$d_1 = A_1 - w_1 - q^n n_2 + n_1, \quad d_2(A_2) = A_2 - w_2 + n_2.$$

Taking the stochastic discount factor as given, the firm's problem is

$$V(n_1, A_1) = \max_{n_2 \geq 0} \left\{ d_1 - C(d_1) + \mathbb{E}[\mathcal{M}(A_2)(d_2(A_2) - C(d_2(A_2)))] \right\}.$$

As in the main text, let $q = \mathbb{E}[\mathcal{M}(A_2)]$, define the low-productivity state $A_2^L = \bar{A} - \Delta_A$, and write $\mathcal{M}^L = \mathcal{M}(A_2^L)$. The cash level that fully insures the firm against equity issuance is

$$n_2^{\max} = w_2 - A_2^L = w_2 + \Delta_A - \bar{A}.$$

At and above $n_2^{\max}$, the firm does not issue equity in either period-2 state. Under the maintained condition $d_1 \geq 0$, the marginal value of additional cash is $q - q^n < 0$. Thus, the firm never chooses cash above $n_2^{\max}$.

For $0 < n_2 < n_2^{\max}$, equity issuance occurs only in the low-productivity state. The first-order condition is

$$-q^n + q + \frac{\mathcal{M}^L}{2} [\mu_1 - \mu_2 d_2(A_2^L)] = 0.$$

Substituting $d_2(A_2^L) = n_2 - n_2^{\max}$ gives

$$n_2^* = \min \left\{ n_2^{\max}, \max \left\{ 0, n_2^{\max} + \frac{\mu_1}{\mu_2} - \frac{2(q^n - q)}{\mu_2 \mathcal{M}^L} \right\} \right\}.$$

Proposition B.1 (Precautionary savings with a linear issuance cost).

Suppose $d_1 \geq 0$ and $q^n > q$. The firm's optimal cash holding is weakly increasing in the proportional issuance-cost parameter μ_1 and is bounded above by $n_2^{\max}$.

Proof. In the interior region,

$$\frac{\partial n_2^*}{\partial \mu_1} = \frac{1}{\mu_2} > 0.$$

At the lower and upper bounds the policy is locally constant, so the comparative static is weak. The upper bound follows because cash above $n_2^{\max}$ no longer reduces issuance costs and earns a lower return than the risk-free asset. ■

C Decreasing differences in the proof of Proposition 4

This appendix establishes the first step in the proof of Proposition 4. In particular, we show that the steady-state Bellman operator preserves concavity in cash and decreasing differences in cash and productivity. Under the maintained conditions of Section 4 — a concave, continuously differentiable current payoff g with g' weakly decreasing and $g' \geq 1$; weakly increasing operating profits $\pi(z)$ on the finite grid $\mathbb{Z}$; a productivity kernel that is weakly increasing in current z in the sense of first-order stochastic dominance (FOSD); and $\beta \in (0, 1)$ — the steady-state value function J is weakly increasing and concave in n and has weakly decreasing differences in (n, z) .

Restrict cash holdings to $n \in [0, N]$ and impose $n' \in [0, N]$ for a large constant N . The truncation is innocuous: it never binds in the computed solution, and none of the steps below depend on N . Let $\mathcal{B}$ denote the space of bounded continuous functions on $[0, N] \times \mathbb{Z}$ with the sup norm, and define the Bellman operator

$$(TJ)(n, z) = \max_{n' \in [0, N]} \{g(\pi(z) + n - q^n n') + \beta \mathbb{E}[J(n', z') \mid z]\}.$$

T satisfies Blackwell's sufficient conditions and is a β -contraction on $\mathcal{B}$. Let $\mathcal{S} \subset \mathcal{B}$ collect the functions that are, for each z , weakly increasing and concave in n , and that have weakly decreasing differences in (n, z) : $J(n_2, z) - J(n_1, z)$ is weakly decreasing in z whenever $n_2 > n_1$. The set $\mathcal{S}$ is nonempty ($J \equiv 0$) and closed under uniform convergence, because monotonicity, concavity, and decreasing differences are preserved by pointwise limits. By the corollary to the contraction mapping theorem, it therefore suffices to show that $T(\mathcal{S}) \subseteq \mathcal{S}$: the unique fixed point then lies in $\mathcal{S}$.

Fix $J \in \mathcal{S}$. Write cash on hand as $m = \pi(z) + n$ and pass to net payout $d = m - q^n n'$ as the choice variable:

$$(TJ)(n, z) = W(\pi(z) + n, z), \quad W(m, z) \equiv \max_{d \in \mathcal{D}(m)} F(d; m, z),$$

where

$$F(d; m, z) \equiv g(d) + \beta \mathbb{E}\left[J\left(\frac{m-d}{q^n}, z'\right) \mid z\right], \quad \mathcal{D}(m) \equiv [m - q^n N, m].$$

Step 1: W is weakly increasing and concave in m . The map $(d, m) \mapsto (m - d)/q^n$ is linear, $J(\cdot, z')$ is concave, and g is concave, so F is jointly concave in (d, m) ; the graph of $\mathcal{D}$ is convex; hence $W(\cdot, z)$ is concave. Monotonicity follows from $g' \geq 1 > 0$ and J weakly increasing, and continuity is standard. Since $m = \pi(z) + n$ is affine and increasing in n ,

TJ inherits boundedness, continuity, monotonicity, and concavity in n .

Step 2: the optimal payout is weakly increasing in m and in z . For $d_2 > d_1$, set $\omega \equiv (d_2 - d_1)/q^n > 0$ and $a \equiv (m - d_2)/q^n$, so that

$$F(d_2; m, z) - F(d_1; m, z) = g(d_2) - g(d_1) - \beta \mathbb{E}[J(a + \omega, z') - J(a, z') \mid z].$$

The bracketed increment has fixed width ω . By concavity of $J(\cdot, z')$ it is weakly decreasing in the location a , and by decreasing differences of J it is weakly decreasing in z' at every location. Therefore: (i) raising m raises a and weakly lowers the increment state by state, so F has weakly increasing differences in (d, m) ; and (ii) raising z shifts the kernel upward in the FOSD order while the increment is weakly decreasing in z' , so the conditional expectation weakly falls and F has weakly increasing differences in (d, z) . Since the feasible set $\mathcal{D}(m)$ ascends with m (both endpoints increase) and does not depend on z , Topkis's monotonicity theorem implies that the extremal optimal selections $d^*(m, z)$ are weakly increasing in m and in z .

Step 3: envelope. In the original formulation the feasible set for n' does not vary with m and g is continuously differentiable, so wherever the concave function $W(\cdot, z)$ is differentiable — almost everywhere — Danskin's theorem gives

$$W_m(m, z) = g'(d^*(m, z)),$$

and the value is the same for every optimal selection. Concavity also makes $W(\cdot, z)$ absolutely continuous, so for $m_2 > m_1$,

$$W(m_2, z) - W(m_1, z) = \int_{m_1}^{m_2} g'(d^*(u, z)) du.$$

Step 4: decreasing differences of TJ . Fix $n_2 > n_1$ and $z_2 \geq z_1$. Using $(TJ)(n, z) = W(\pi(z) + n, z)$ and Step 3,

$$(TJ)(n_2, z_i) - (TJ)(n_1, z_i) = \int_{n_1}^{n_2} g'(d^*(\pi(z_i) + u, z_i)) du, \quad i \in \{1, 2\}.$$

Since $\pi(z_2) \geq \pi(z_1)$ and d^* is weakly increasing in both arguments (Step 2), $d^*(\pi(z_2) + u, z_2) \geq d^*(\pi(z_1) + u, z_1)$ for every u ; since g' is weakly decreasing, the integrand under z_2 is pointwise weakly smaller. Hence

$$(TJ)(n_2, z_2) - (TJ)(n_1, z_2) \leq (TJ)(n_2, z_1) - (TJ)(n_1, z_1),$$

so TJ has weakly decreasing differences in (n, z) . Together with Step 1, $TJ \in \mathcal{S}$, which completes the invariance argument, and the fixed point J inherits every property defining $\mathcal{S}$. ■

Two remarks. First, for $n'_2 > n'_1$, decreasing differences of J and the FOSD kernel imply that $\mathbb{E}[J(n'_2, z') - J(n'_1, z') \mid z]$ is weakly decreasing in current z , which is precisely the form used in the proof of Proposition 4; wherever the derivative exists, $M(n', z) = \beta \mathbb{E}[J_n(n', z') \mid z]$ is weakly decreasing in n' (by concavity) and in z . Second, the argument delivers the weak inequalities only; the strict monotonicity invoked for the strict decline of the target remains a property of the calibrated solution, in which the decreasing-differences inequality holds strictly up to solver tolerance. The FOSD ordering of the discretized kernel is verified directly for the calibrated transition matrix: the conditional distribution functions are ordered cell by cell, with no violations.

D Algorithm - Repeated transition method

This section explains how we compute the recursive competitive equilibrium and why the aggregate dynamics generated by the model cannot be summarized well by a log-linear law of motion in the contemporaneous aggregate cash stock alone. The key computational challenge is that firm-level cash policies are highly nonlinear and occasionally constrained, while the aggregate equilibrium also requires market clearing in labor and consumption goods. We therefore use the repeated transition method that does not impose a parametric aggregate law of motion during the solution. After solving the model, we use the converged simulated allocations to recover and validate the nonlinear aggregate law of motion that the equilibrium implies.

D.1 Repeated transition algorithm

The model is solved using the global nonlinear sequence-space method developed by [Lee \(2026\)](#). The algorithm works on a long simulated path of aggregate productivity shocks and iterates backward over that path. At each iteration, firms solve their dynamic problem taking as given the current guess for the aggregate sequences. Conditional expectations are computed using the value functions from the previous iteration, rather than by explicitly specifying and estimating a finite-dimensional law of motion for aggregate states.

This approach is useful in our environment for three reasons. First, the individual cash policy is nonlinear and contains occasionally binding constraints. Second, the model has non-trivial market-clearing conditions for both labor and the consumption good. Third, the aggregate cash stock is informative about the cross-section, but it is not sufficient to make the true law of motion linear. The repeated-transition method handles these issues by tracing sequences of implied aggregate prices and quantities. In each iteration, the implied price sequences need not clear markets exactly; market clearing is achieved in the limit, when the guessed and realized sequences converge.

This distinction between implied prices and market-clearing prices is important. A standard Krusell-Smith-style algorithm would require an additional loop to clear the non-trivial price p_t in every iteration. The repeated-transition method avoids this inner loop and therefore substantially reduces the computational burden. In the converged solution, the predicted path from the current iteration and the realized path from the next iteration are indistinguishable, so the method delivers the same equilibrium object without imposing log-linearity.

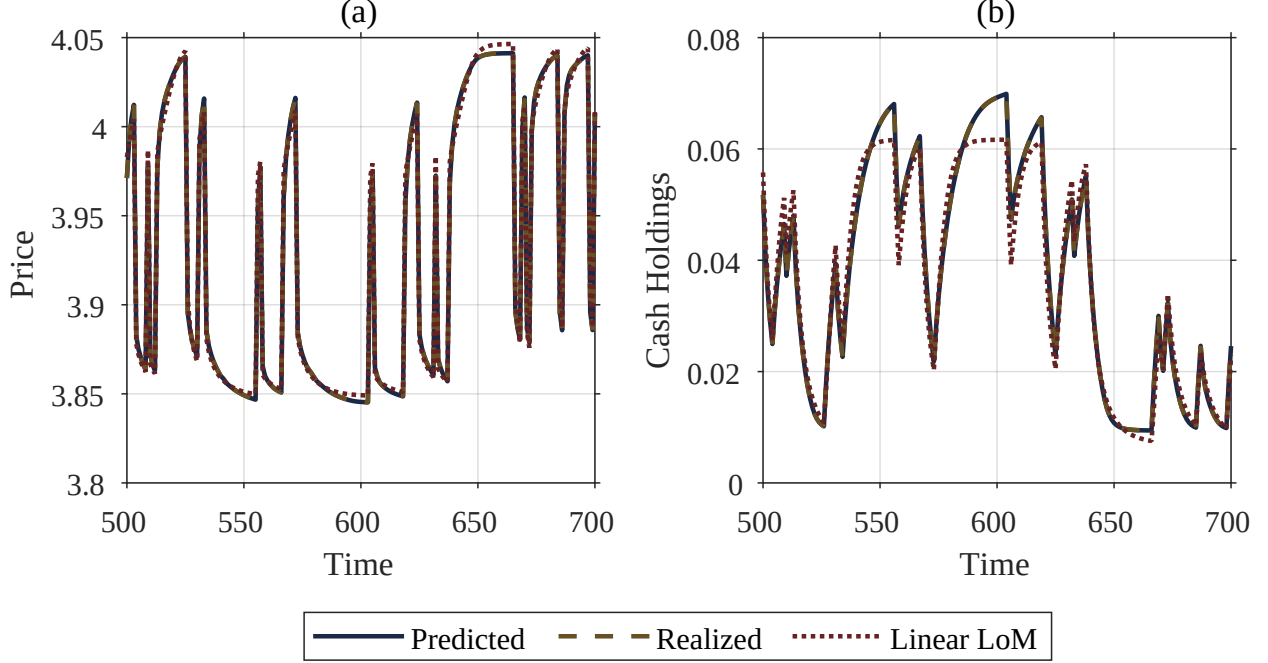


Figure D1: Aggregate fluctuations in the baseline model

Notes: The figure plots the time series of the price p_t and the aggregate cash stock N_t in the baseline model. In both panels, the solid line is the predicted time series (n^{th} guess) $\{p_t^{(n)}, N_t^{(n)}\}_{t=500}^{700}$; the dashed line is the realized time series $\{p_t^*, N_t^*\}_{t=500}^{700}$; the dash-dotted line is the time series implied by the fitted log-linear law of motion.

D.2 Nonlinear aggregate dynamics and the log-linear benchmark

Figure D1 illustrates the aggregate dynamics in the baseline model. The figure compares the converged repeated-transition allocation with the path implied by a log-linear forecasting rule for the aggregate cash stock N_t and the price p_t . The repeated-transition method produces near-perfect alignment between the predicted and realized equilibrium paths, while the log-linear benchmark misses part of the nonlinear adjustment.

To construct the benchmark, we fit the converged repeated-transition allocations using shock-specific log-linear rules:

$$\begin{aligned}
 \log(N_{t+1}) &= -0.5613 + 0.8854 \log(N_t), & \text{if } A_t = A_B, & \quad R^2 = 0.9973, \quad \text{MSE} = 0.0007, \\
 \log(N_{t+1}) &= -0.7794 + 0.7194 \log(N_t), & \text{if } A_t = A_G, & \quad R^2 = 0.9908, \quad \text{MSE} = 0.0013, \\
 \log(p_t) &= 1.3228 - 0.0090 \log(N_t), & \text{if } A_t = A_B, & \quad R^2 = 0.9347, \quad \text{MSE} = 0.0000, \\
 \log(p_t) &= 1.2966 - 0.0071 \log(N_t), & \text{if } A_t = A_G, & \quad R^2 = 0.9594, \quad \text{MSE} = 0.0000.
 \end{aligned}$$

Although the log-linear rule captures part of the movement in N_t , it performs worse for prices and does not reproduce the full equilibrium path. The reason is structural. At the micro level, firms' cash policies become flat once firms approach their target cash posi-

tions, so the distribution does not move linearly with aggregate cash. At the aggregate level, firms cannot trade cash with one another and the wedge between the return on cash and the risk-free bond is fixed by λ . As a result, general equilibrium prices do not smooth cross-sectional cash demand as strongly as in models with a freely adjusting intertemporal price. The log-linear rule therefore misses the history dependence created by the interaction between nonlinear firm policies and the segmented market for cash.

D.3 Recovering the nonlinear law of motion

The repeated-transition method does not require an assumed aggregate law of motion to solve the model. However, after convergence, we can recover the law of motion implied by the equilibrium simulated path. We consider specifications of the form

$$\begin{aligned} N_{t+1} &= G_N(N_t, N_{t-1}, N_{t-2}, \dots, N_{t-n}; A_t), \\ p_t &= G_p(N_t, N_{t-1}, N_{t-2}, \dots, N_{t-n}; A_t). \end{aligned}$$

The first set of specifications uses only contemporaneous aggregate cash N_t , allowing for increasingly high-order polynomial terms. The second set adds lagged aggregate cash stocks, again allowing for nonlinear terms. This exercise separates two questions: whether nonlinear terms in current cash are enough, and whether the equilibrium also contains genuine history dependence.

Table D1 shows that nonlinear terms in current aggregate cash improve the fit, especially for N_{t+1} , but they are not sufficient to recover the equilibrium law of motion. The price rule G_p in particular improves only once lagged aggregate cash terms are included. Allowing the law of motion to depend on up to seven lags of N_t recovers the equilibrium dynamics with an essentially perfect fit. This result clarifies the role of aggregate cash in the repeated-transition algorithm. The current aggregate cash stock is a useful summary for indexing the cross-section along the simulated path, but the true law of motion is richer than a function of current cash alone.

D.4 Out-of-sample validation

We validate the recovered law of motion on an out-of-sample simulated path. We solve the model again using the repeated-transition method on a new aggregate shock sequence and compare the converged equilibrium path with two approximations: the nonlinear recovered law of motion estimated on the original path, and the log-linear rule above.

Table D1: The fit of the law of motion across different specifications

	No. of lags	order	Goodness of fit: R^2			
			$N_{t+1} : \text{Good}$	$N_{t+1} : \text{Bad}$	$p_t : \text{Good}$	$p_t : \text{Bad}$
Contemp.	0	1	0.8956	0.9452	0.9922	0.9966
	0	2	0.9839	0.9952	0.9927	0.9976
	0	3	0.9973	0.9995	0.9930	0.9976
	0	4	0.9993	0.9999	0.9932	0.9976
	0	5	0.9996	1.0000	0.9933	0.9976
Add. history	1	3	0.9999	1.0000	0.9987	0.9979
	2	3	0.9999	1.0000	0.9997	0.9984
	3	3	0.9999	1.0000	0.9998	0.9987
	4	3	0.9999	1.0000	0.9998	0.9991
	5	3	0.9999	1.0000	0.9998	0.9994
	6	3	0.9999	1.0000	0.9998	0.9996
	7	3	0.9999	1.0000	0.9998	0.9997

Notes: The first column (# of lagged) reports the number of lagged terms included in the specification. The second column (order) reports the highest order of polynomials considered. When lagged terms are included, contemporaneous terms are considered up to the fifth polynomial for all specifications. The third and fourth columns reports the R^2 of the law of motion of N_{t+1} for Good and Bad shock realizations. The fifth and sixth columns reports the R^2 of the law of motion of p_t for Good and Bad shock realizations.

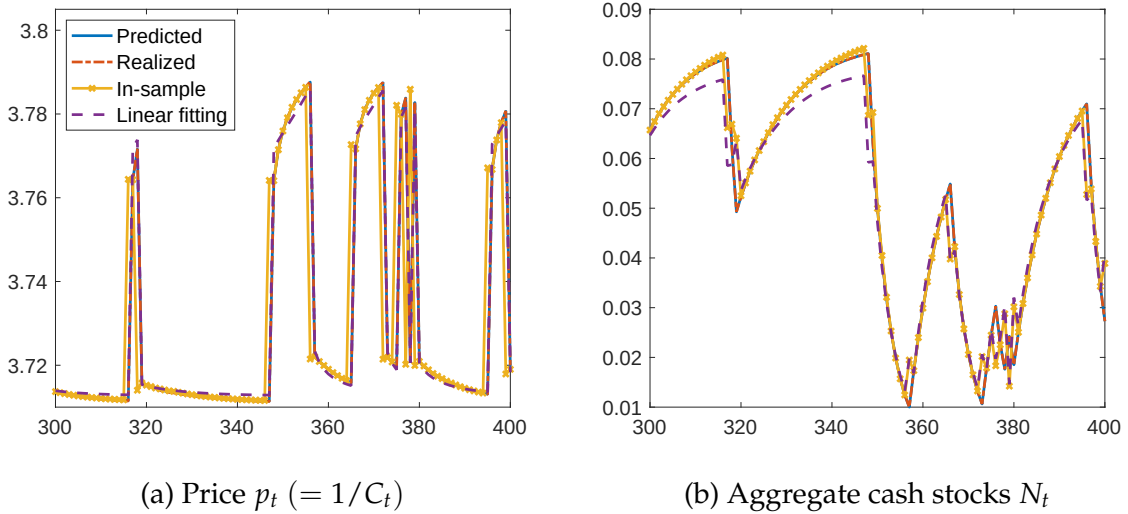


Figure D2: Fit on the out-of-sample path

Notes: The figure plots the time series of the price p_t and the aggregate cash stock N_t in the baseline model for an out-of-sample simulated path of the aggregate shocks. In both panels, the solid blue line is the predicted time series (n^{th} guess) $\{p_t^{(n)}, N_t^{(n)}\}_{t=300}^{400}$; the orange dashed line is the realized time series $\{p_t^*, N_t^*\}_{t=300}^{400}$; the yellow line with x-markers is the in-sample prediction from the exact law of motion recovered from the in-sample path; and the purple dashed line is the prediction implied by the linear law of motion.

Figure D2 shows that the recovered nonlinear law of motion tracks the out-of-sample

equilibrium path closely for both p_t and N_t . The R^2 of the recovered rule is above 0.999 for both objects in both aggregate states. In contrast, the log-linear rule misses part of the adjustment, especially for prices. The main conclusion is therefore not only that the model generates nonlinear aggregate fluctuations, but also that this nonlinearity is quantitatively relevant for the equilibrium law of motion. The repeated-transition method is well suited to this environment because it solves the nonlinear equilibrium directly and lets us recover the richer law of motion ex post, rather than imposing a log-linear forecasting rule during the solution.

D.5 Note on equilibrium price

The equilibrium price p_t is determined from the following variant of the non-trivial market clearing condition:¹

$$p = \arg_{\tilde{p}} \left\{ c(\tilde{p}) - \int [d(x; X, \tilde{p}) + w(X, \tilde{p})l(x; X, \tilde{p})] d\Phi = 0 \right\}.$$

This is a fixed-point problem and computationally costly to solve. Instead, the repeated transition method uses the implied price p^* , which is obtained as follows:

$$\begin{aligned} p^* &= \arg_{\tilde{p}} \left\{ c(\tilde{p}) - \int [d(x; X, p^{(n)}) + w(X, p^{(n)})l(x; X, p^{(n)})] d\Phi = 0 \right\} \\ &= 1 / \int [d(x; X, p^{(n)}) + w(X, p^{(n)})l(x; X, p^{(n)})] d\Phi, \end{aligned}$$

where $p^{(n)}$ is the guessed price in the n^{th} iteration. We take a sufficient-statistic approach, using the aggregate cash stock N_t , the first moment of the distribution of cash holdings, as the sufficient statistic.

¹This condition is derived from combining the household's budget constraint and the equity market clearing condition. For a sharp illustration, the time subscript is lifted.

E Additional model results

E.1 Representative firm model

Figure E1 plots the GIRF to a negative TFP shock when cash is at median and low levels over the business cycle. Panel (a) plots the GIRF for the baseline model with heterogeneous firms. Panel (b) plots the GIRF for the representative-firm model, which has aggregate TFP shocks but no idiosyncratic TFP shocks. The figure illustrates that the nonlinearity, while it is still present in the representative firm model, is severely amplified by firm heterogeneity: the gap between the low-cash and median-cash consumption responses—our measure of state dependence—is about three times larger in the heterogeneous-firm model. This comes from the fact that the representative firm is rarely at the two extremes of the cash policy function, where the nonlinearity is stronger, resulting in an almost linear business cycle.

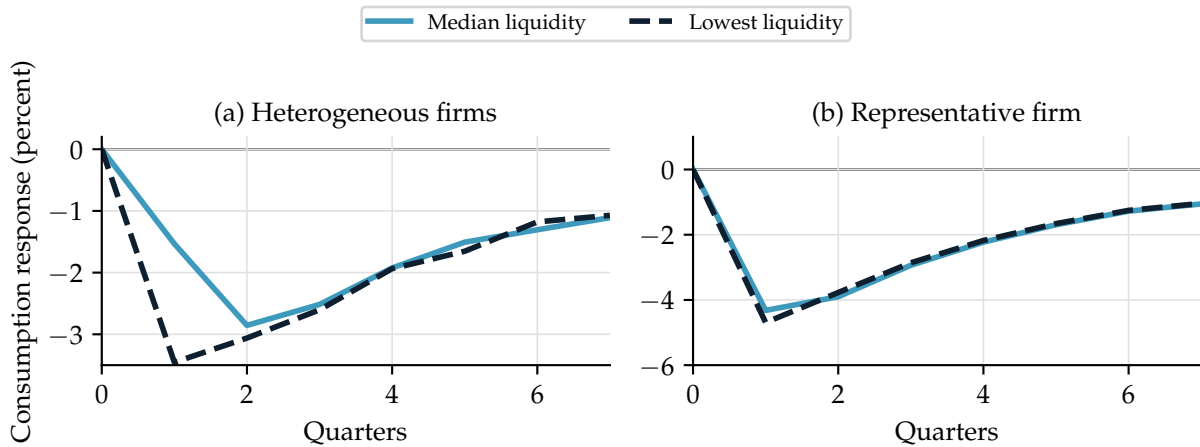


Figure E1: GIRF: Consumption responses to negative shocks for median and low cash in the baseline model (panel a) and the representative firm model (panel b).

Notes: The figure plots the mean percentage response of consumption to a negative aggregate TFP shock over the first eight quarters, relative to matched no-shock simulations. The light-blue solid line starts from median liquidity and the dark-blue dashed line starts from the lowest-liquidity state.

E.2 Uncertainty shocks

We subject the model to a Bloom-type aggregate uncertainty shock. Uncertainty is a two-state process, $S_t \in \{L, H\}$, whose quarterly transition matrix follows Bloom et al. (2018). We hold this transition matrix fixed across the two uncertainty regimes, so that the experiment changes the dispersion of future aggregate productivity without changing the

persistence or mean transition structure of the process.

Conditional on the low-uncertainty state, next period’s aggregate TFP takes values on the grid $A \in \{0.980, 1.020\}$. In the high-uncertainty state, we widen this grid by a factor of 1.6, so that $A \in \{0.968, 1.032\}$. The midpoint remains unchanged in both regimes. Thus, this is a *pure second-moment shock*: on impact the level of aggregate TFP is unchanged, while the dispersion of next period’s TFP increases in the high-uncertainty state. The experiment therefore asks whether the model’s precautionary cash mechanism responds to uncertainty in the same way that it responds to first-moment shocks.

Table E1 summarizes the model’s behavior conditional on the realized uncertainty state. The comparison focuses on three implications of the mechanism: whether firms accumulate more cash, whether they retain more of a marginal dollar of liquidity, and whether the greater dispersion of aggregate productivity translates into more volatile output and consumption.

Table E1: Business-cycle statistics by uncertainty state

	low uncertainty ($S = L$)	high uncertainty ($S = H$)
share of periods	0.667	0.333
mean aggregate cash	0.0358	0.0414
mean aggregate MPPO	0.829	0.805
std log(output)	0.0216	0.0347
std log(consumption)	0.0235	0.0322

Notes: Statistics are computed over the simulated path, conditioning on the realized uncertainty state S_t . Means of output and consumption are essentially equal across states (a pure second-moment shock moves the variances, not the means). Cash and output/consumption volatilities are aggregate objects and are robust to the numerical solution tolerance.

The results confirm all three implications. In high-uncertainty periods, firms hold more cash: mean aggregate cash rises from 0.0358 to 0.0414. They also retain more of the marginal dollar, as the mean aggregate MPPO falls from 0.829 to 0.805. This is precautionary hoarding responding to uncertainty — the general-equilibrium counterpart of the finance-uncertainty multiplier of [Alfaro et al. \(2024\)](#), operating through the paper’s non-monotone cash policy and the MPPO. At the same time, output and consumption become more volatile, while their means remain essentially unchanged. In short, uncertainty raises corporate liquidity demand, lowers the propensity to pay out, and amplifies aggregate volatility. The state-dependence result at the heart of the paper therefore extends to uncertainty-driven fluctuations.

F A structural interpretation of the cash wedge via a reserve requirement

F.1 Environment

This appendix gives the cash wedge a *structural interpretation*, in the spirit of Jeenas (2023), who intermediates firms' liquid-asset holdings through a competitive financial intermediary in a heterogeneous-firm model with financial constraints. The below-market return on corporate cash arises as the pass-through of a reserve requirement: the competitive intermediary that issues deposits must back them with central-bank reserves remunerated below the market rate, and passes that cost through to the deposit return. The one substantive difference from Jeenas is what becomes of the intermediation cost: he rebates it lump-sum to households, so it nets out of the resource constraint, whereas here it is a genuine resource cost.

Two institutions and one technology are added behind the market for cash: (i) a *competitive financial intermediary*, which issues the deposits that firms hold as cash; (ii) a *central bank*, which issues the reserves that back those deposits and sets their remuneration; and (iii) a one-period *storage technology*, operated by the central bank, which carries the backing from issuance to maturity. Households, firms, and the government are otherwise unchanged. The three paragraphs below describe, in turn, the deposit instrument, the regulation linking deposits to reserves, and the settlement.

Deposits “Cash” is a one-period deposit issued by a competitive financial intermediary: one unit bought at date t at price q_t^n pays one unit of the final good at $t+1$. The firm's budget constraint is identical to the main text, $d = \pi(z; X) + n - q_t^n n' + T^f(X)$ (with $T^f = 0$ in the baseline), with $n' \geq 0$.

Reserve requirement The intermediary operates at zero cost but faces a regulatory requirement: every unit of deposit face value must be backed one-for-one by central-bank reserves. A reserve is a claim sold by the central bank at t at price q_t^r that pays one unit of the good at $t+1$. Reserves are remunerated *below market* by policy:

$$1 + r_t^r = 1 + \lambda r_t, \quad 0 < \lambda < 1, \quad q_t^r = \frac{1}{1 + \lambda r_t},$$

where r_t is the equilibrium risk-free rate ($q_t = 1/(1 + r_t)$). The single parameter $\lambda \in (0, 1)$ is the fraction of the market rate that remunerated reserves earn: reserves pay $r_t^r = \lambda r_t$,

below the risk-free rate. Its calibrated value $\lambda = 0.87$ says reserves earn 87% of the market rate, an interest-on-reserves rate set below the policy rate, as central banks do in practice. Competition passes this shortfall straight through to the deposit return, so λ is equally the wedge on cash, $r_t^n = \lambda r_t$. The requirement is an institutional constraint (liquidity regulation).

Settlement The reserve proceeds $q_t^r N_t'$ that the central bank receives at issuance ($N_t' \equiv \int n' d\Phi_t$) are placed in a one-period storage technology and redeemed to cover the maturing deposits $N_{t+1} = N_t'$ at $t+1$. Since $q_t^n = q_t^r$, what the technology carries forward is exactly the firm's payment for cash: it absorbs $q_t^n N_t'$ today and returns face N_t' next period, a below-market return $r_t^n = \lambda r_t$. The goods the firm gives up are deferred, not rebated, so the wedge is a real cost rather than a pure transfer, and *no tax on households is required*.

F.2 Equilibrium wedge

Zero profit of the competitive intermediary, period by period, requires the deposit price to equal the cost of the mandated backing:

$$q_t^n = q_t^r = \frac{1}{1 + \lambda r_t} \iff r_t^n = \lambda r_t.$$

With $\lambda = 0.87$, the net return on cash is 87% of the risk-free rate, exactly the [Cooley and Quadrini \(2001\)](#) wedge used in the calibration, and *exactly* the pricing rule implemented in the computation ($r_t^n = 0.87 r_t$ each period). The requirement binds: reserves are return-dominated by bonds for any $\lambda < 1$, so the intermediary holds no excess reserves; and deposits, being return-dominated in turn, are held by firms only for the liquidity role given to them by the $n' \geq 0$ constraint and the issuance cost. The deposit supply is perfectly elastic at the policy-determined wedge, so the market-clearing condition for liquid assets is trivially satisfied at any $\int n' d\Phi$. This is precisely the main text's $\zeta \rightarrow 0$ assumption of an infinitely elastic net supply of cash (under which the market for cash always clears without solving for a spot rate), which this appendix thereby interprets.²

²An equivalent parameterization uses a *partial* requirement: a fraction α_r of deposit *face value* must be backed by unremunerated reserves ($\bar{r}^r = 0$) and the remainder by government bills priced at q_t , both carried by the storage technology. Zero profit then gives $q_t^n = \alpha_r + (1 - \alpha_r)q_t$, i.e. $r_t^n = (1 - \alpha_r)r_t / (1 + \alpha_r r_t) \approx (1 - \alpha_r)r_t$, so $\alpha_r = 1 - \lambda = 0.13$ replicates the same wedge to first order. This variant puts government bills in positive net supply (held by the intermediary), so the household bond market-clearing condition and the government budget must be restated accordingly.

F.3 Aggregation

Aggregate net payouts are $D_t = \Pi_t + N_t - q_t^n N_t'$ with $\Pi_t = Y_t - w_t L_t - \xi$. The household budget at t reads $C_t = w_t L_t + D_t$ (bonds are in zero net supply; equity positions net out), so

$$C_t = Y_t - \xi - \underbrace{(q_t^n N_t' - N_t)}_{H_t = \text{net goods flow into storage}}. \quad (\text{F.1})$$

This is exactly the goods market of the business-cycle model: net accumulation of corporate cash absorbs final goods at the price q_t^n the firm pays. Cash matters for aggregates only insofar as it carries a real cost: in a frictionless economy it would be a pure financial claim, $H_t \equiv 0$, and its accumulation neutral.³ Charging the full price q_t^n as a resource cost is the maximal such case, which we adopt for concreteness and for consistency with the reduced-form business-cycle model; the state dependence survives for any positive cost, not only this one. Three observations:

1. **Impact:** A hoarding surge at t ($N_t' \uparrow$) lowers C_t : the firm pays q_t^n per unit of new cash, so consumption falls by $q_t^n N_t' - N_t$ net of the maturing stock. This is an *impact* effect: the resources return at maturity (next point), so the present-value loss per unit is only $q_t^n - q_t$, the wedge.
2. **Deferred return:** The stored resources flow back at maturity: what is placed in storage at t is redeemed as $N_{t+1} = N_t'$ at $t+1$, and C rises above $Y - \xi$ exactly when firms decumulate ($q_t^n N_t' < N_t$). The mechanism is therefore an *intertemporal timing* effect under a frictional wedge, not wealth destruction: the impact response is intact even though the resources are recovered. This is the same logic as the main text's fiscal-transfer experiment, where a PV-neutral transfer has real effects because payout timing matters when the MPPO is below one.
3. **No equity offset:** The representative household holds all equity and cannot change its net position in equilibrium ($s' = s$), so the equity price appears on both sides of its budget constraint and drops out: capital gains are a valuation, not an aggregate resource. Only a *contemporaneous* rebate of the firm's payment could offset the dividend cut, and deferring it into storage rules that out.

³Previous studies, including [Covas and Driscoll \(2014\)](#); [Van den Heuvel \(2022\)](#), show that liquidity requirements can impose real economic costs by reducing credit provision and diverting resources from more productive uses.

Aggregate state The goods market (F.1) involves only current-period objects, so the aggregate state is unchanged. Both the outflow $q_t^n N_t'$ and the redemption $N_t = \int n d\Phi_t$ are functions of $\{\Phi_t, A_t\}$ and the current price q_t^n (itself a function of the state), and the storage technology self-finances: what it carries from $t - 1$ earns exactly the return r^n needed to redeem the maturing N_t , so it demands no external funding. No quantity or price dated $t - 1$ enters any date- t budget, so the state remains $X_t = \{\Phi_t, A_t\}$ and the existing solution algorithm, written for $\{\Phi, A\}$, carries over unchanged.

Knife-edge closure What keeps the wedge real is that the firm's payment is deferred into storage, not returned to households within the period. Were the proceeds $q_t^n N_t'$ instead rebated to households at issuance, then $H_t \equiv 0$ in (F.1) and $C_t = Y_t - \xi$ for *any* path of $\{N_t'\}$: cash would be a pure asset swap with no real effects, eliminating the paper's positive results, since the state-dependent consumption responses and the fiscal-policy analysis all run through H_t . Deferring the payment one period, rather than rebating it, is what keeps the cash margin real.

F.4 Two-period version

In the two-period model the firm buys n_2 deposits at price $q^n = q^r$ in period 1, and the storage technology carries the payment $q^n n_2$ forward to redeem n_2 in period 2. Household consumption is

$$c_1 = A_1 + n_1 - q^n n_2, \quad \tilde{c}_2 = \tilde{A}_2 + n_2,$$

exactly as in the reduced-form model in the main text: the firm's net payout falls by $q^n n_2$, no household tax is levied, and the cash return is $r^n = \lambda r$. The firm hoards cash as it carries a liquidity value, and all main-text propositions turn on that inequality unchanged.

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